

Fundamentals of Machine learning for and with engineering applications

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1 Recaps: Uncertainty, Probability and Models

2 Descriptive and Predictive statistics

3 Simulations

4 Data properties

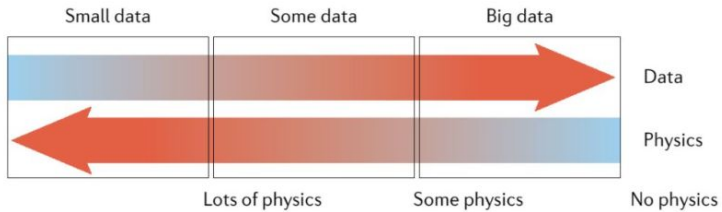
5 Data risk and bias

6 Metrics and distributions

7 Information Theory

8 Comparing information

Data vs Physics



- Def 1: Not knowing if an event is true or false. (Useful)
- Def 2: Things that cannot be measured. (Not useful)

Probability is how Uncertainty is quantified!

- Clarity test
- Assign a number between 0 and 1 to our degree of belief
- Error definition

Sentence also good for fortune cookies

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What is Statistics

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Estimation

- Process of obtaining the best value or range of a property in an unsampled location
- Local accuracy takes precedence over global spatial variability
- Not appropriate for forecasting

Inference

- Predict unseen samples given assumptions about the population
- Test with a pre-trained model (ML definition)
- Generality versus Accuracy

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Variables and Features, Labels and Instances

Population

Exhaustive, finite list of properties of interest over area of interest.

Generally the entire population is not accessible

Samples/experiments/instances

The set of values and location that have been measured.

How many experiments are needed?

Sampling distribution -(IID ?)

identical and independent distribution The set of values and location that have been measured.

Features

The values to be measured for each sample/experiment/instance.

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Predictors = input variables, $X_1, \dots, X_M$

Response = output variables

Error

Deviation from ... exact value (or expected value, mean value, trend...?)

Errors without definitions are just numbers.

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Values, without error, are just number!!

Predictor and Response Features

Given a model $Y = f(X_1, \dots, X_M) + e$

!Here and error! But is it even an error?

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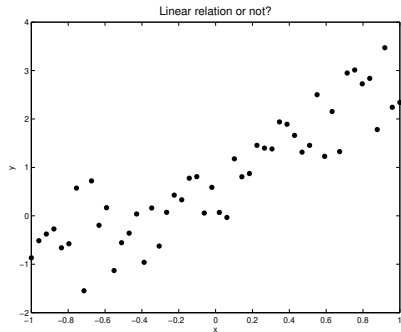
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Soft modeling is in most cases based on **multivariate statistical methods**. Many of these methods may be viewed as sophisticated ways of performing curve fitting to data.



What would be the best model?

- Straight?: $y(x) = ax + b$
- Parabolic?: $y(x) = ax^2 + bx + c$
- Trigonometric?:
 $y(x) = a\sin(x) + b\cos(x)$

Given a model, Generate multiple simulation to represent uncertainty

- Realizations: for the same input parameters, different random numbers.
- Scenarios: different input parameters.

Sampling representative.

Random sampling

Each item of the population has an equal chance of being chosen.

- Very expensive
- Mostly not interesting
- Gives some global properties

Bias sampling

Selection of data is (arbitrarily) distorted

- Sample probability bias has to be corrected for
- Might not capture the global picture
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- anchoring: The first bits are over-considered
- availability: over-estimating the importance of info
- bandwagon: P increases with the number of people holding a belief
- blind spot: not seen biases
- choice supporting: commitment/decision dependent
- clustering illusion: seeing patterns in random events
- confirmation bias
- conservatism bias
- Recency bias
- Supervision bias
- Many many more!

Bias DO NOT cancel out! They sum up (or multiply?)

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Process of obtaining one or more values of a property

- Improved Global accuracy
- Better property distributions

Why simulations then?

- We need to capture the full distribution of properties, extremes matter!
- We need more realistic models.

Why not?

- High dimensionality level
- Computationally expensive
- Convergence limitations
- Constitutive equations need to be rather accurate.

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DATA



SORTED



ARRANGED



PRESENTED
VISUALLY



EXPLAINED
WITH A STORY



- Categorical / Nominal (classes)
- Categorical / Ordinal
- Continuous / Interval (e.g. Celsius)
- Continuous / Ratio
- Discrete: binned/grouped data
- Hard data: direct measurements
- Soft data: indirect measurements, very uncertain
- Primary data: variable(s) of interest
- Secondary data: descriptors
- Collective variables
- Latent variables

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A representation should **capture** the nature of the subject being studied.

Example: If you want to evaluate the 3D structure of a wind turbine, a set of descriptors can be:

- 1 Blade length
- 2 Turbine height
- 3 Geographical position
- 4 Output power
- 5 Wind direction

which are two decimal numbers, a 2d tuple, a 1D time series and a 2D time series (or 3D even).

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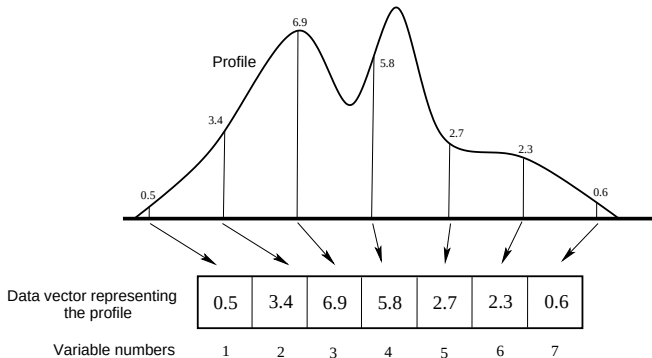
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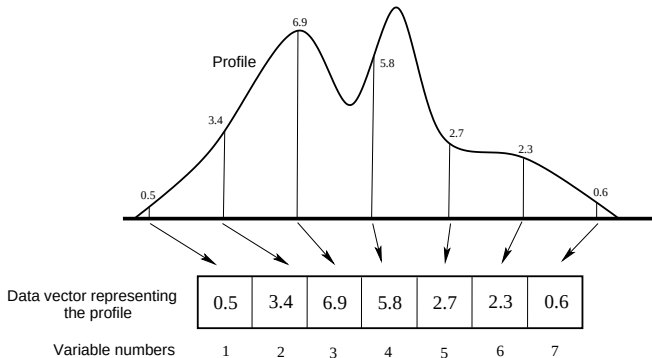
Sampling point representation (SPR)

- An intuitive way to represent curves and spectra is the **sampling point representation**.
- We sample at regular intervals where each sample point is represented by a variable



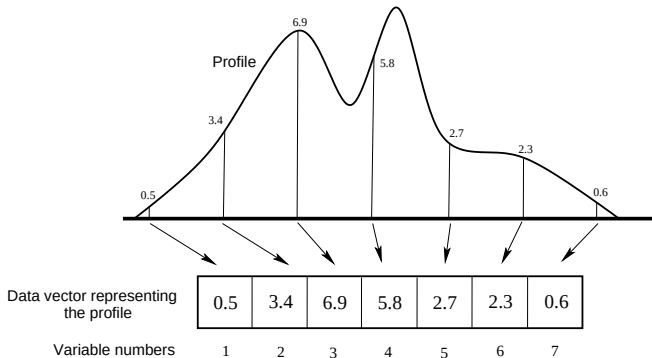
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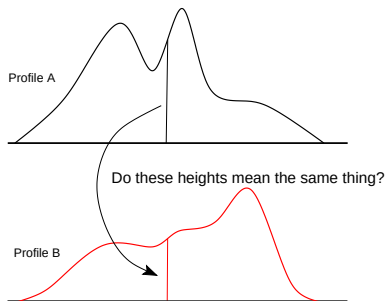
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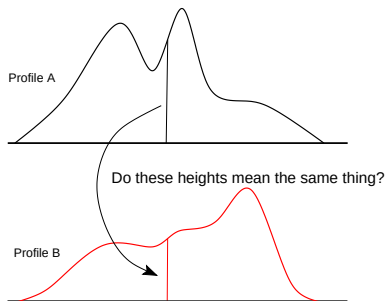
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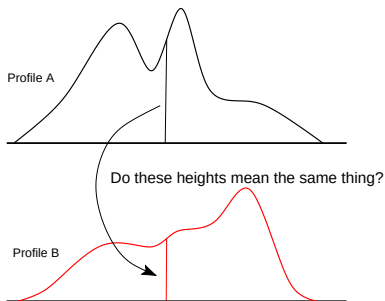
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Discussion point!

How do we compare two wind turbines accounting for the 5 variables previously introduced?

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- All starts from data: what are data-properties?
- Are there such things as good data and bad data?

Life lesson (or exam question, same thing ;))

- Data **DO NOT** always have value.

- TRASH in TRASH out

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Given a representation, it is then needed to decide on a suitable **data structure** for the problem.

Definition

A data structure is a way of storing and organising data in a computer so that it can be used effectively.

Typical data structures used in data analysis are:

- Data points
- Arrays (vectors, matrices, N-mode (way) arrays)
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Data has to be prepared with these steps in mind

- ① Plan experiments: Use experimental design to set up experiments in a *systematic* way
- ② Pre-processing: Is there systematic variation in the data which should be removed Can cross-checking/validation procedures be designed?
- ③ Examine the data: Look at data (tables and plots). Strange behaviours? Smooth behaviour? WARNING!
- ④ Define desired model outcomes (speed, accuracy, false positive/negatives rate)
- ⑤ Estimate and validate model: What do the results tell us? Is the generated model general (valid for future sampling)?
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Data dimensionality

1 D

logs

2 D: maps

Quite limited but great for visualization

3 D

3d maps, seismic cubes. More informative, mostly ok in digital formats.

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Trajectories

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Statistics is collecting, organising, and interpreting data

Spatial and temporal statistics is a branch of applied statistics that emphasises:

- ① the geo context of the data
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The data matrix is an extremely common data structure.

$$X = \begin{bmatrix} 95 & 89 & 82 \\ 23 & 76 & 44 \\ 61 & 46 & 62 \\ 49 & 2 & 79 \end{bmatrix}$$

In python these can be saved as

- lists (vanilla python)
- `numpy.array`s
- pandas dataframes

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There are different conventions. Commonly we will construct data matrix such that:

- Rows are called instances, objects or samples.
- Columns are called features, variables.

One can think of each row to be an experiment, and the rows its properties. Each row (experiment, object, sample, ...) is thus a list of values, one for property.

Note:

Mathematically speaking, this is just a notation. As long as one keeps track and is consistent, columns can be used as rows and vice versa.

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Environmental measurements of rivers. Features (properties):

- pH
- Temperature
- Concentration of pollutants
- Flow rate
- water speed

The instances/experiments/observations/sample:

- Po
- Danube
- Rio delle Amazzoni
- Sjoa
- Atna

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- Flow rate
- water speed

The instances/experiments/observations/sample:

- Po
- Danube
- Rio delle Amazzoni
- Sjoa
- Atna

Different measurements can lead to different data distribution and numerical values.

When fed to numerical recipes, data is just a set of numbers.

For faster computations, data can be **normalized**: i.e. rescaled to the same numerical interval.

Normalisation

The procedure simplifies the numerical recipes work, but it also loses information

As it is not possible to complete sample any distribution, data might be disproportionately handled depending upon data collection (i.e. this can be an error source).

1 Recaps: Uncertainty, Probability and Models

2 Descriptive and Predictive statistics

3 Simulations

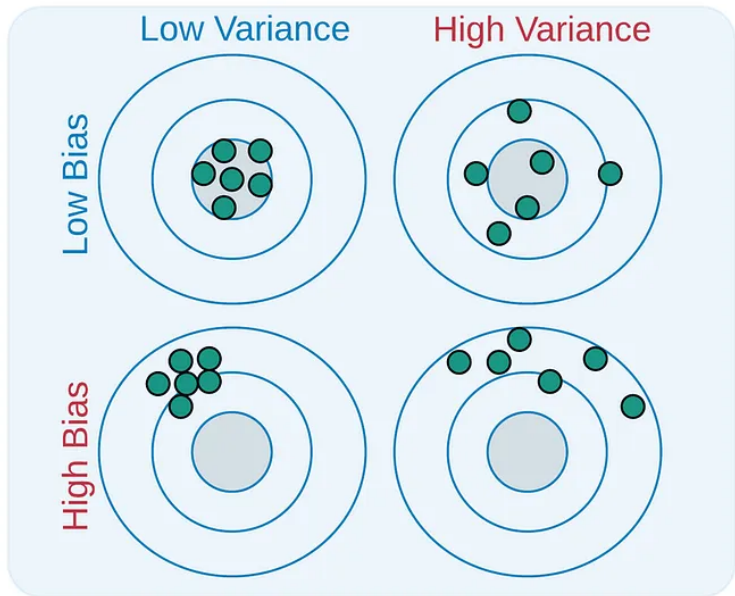
4 Data properties

5 Data risk and bias

6 Metrics and distributions

7 Information Theory

8 Comparing information



- Clarity test

project

- Measured or asserted Probability ?

project

- Clarity test

project

- Measured or asserted Probability ?

project

- Risk = probability * consequences
- Value of information

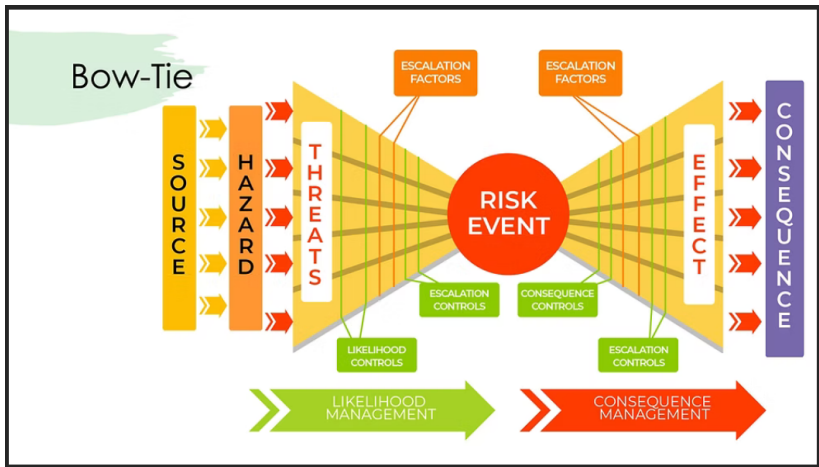
project

- Risk attitude

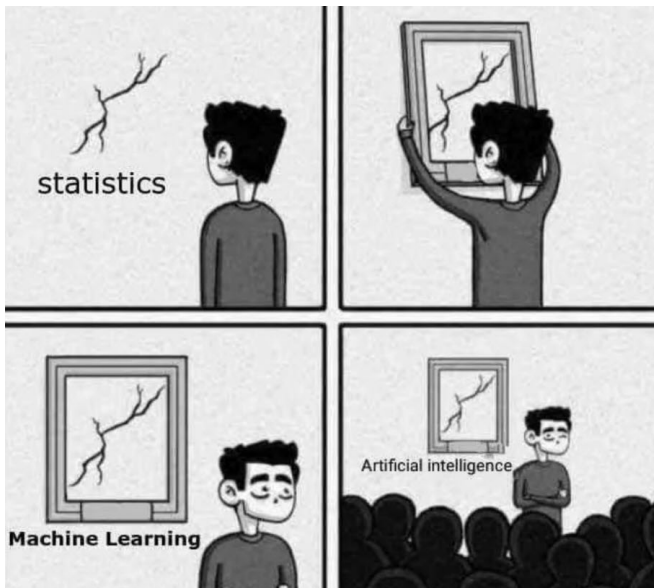
project

Vibe coding, what possibly can go wrong?





A bias recap



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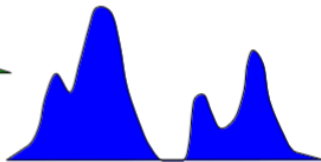
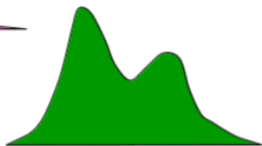
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A metric is needed

Amount of uncertainty

Number of Modes: unimodal, bimodal, polymodal

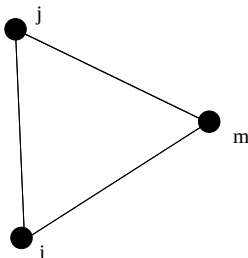


Triangle inequality

Considering a vectors in an N-dimensional space, to be a **distance** it must satisfy the **triangle inequality**:

$$d_{ij} + d_{im} \geq d_{jm}$$

If also $d_{jj} = 0$, if $d_{ij} - d_{ji} = 0$, then we call it a *metric*.



Common metrics:

- Counting
- Euclidean

$$d_{ij}^{(E)} = \left[\sum_{k=1}^N (x_{ik} - x_{jk})^2 \right]^{\frac{1}{2}}$$

Categorical variables do not necessarily satisfy these relations.

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Code a discrete probability distribution in Python

Calculate the Mean and Standard Deviation

How to get an experiment out of this distribution?

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First, the origins:

- Originator: Claude Shannon (1940)
- His initial work has been done on signal transmission.
- It uses Entropy as key measurement of information uncertainty.

How to relate a distribution to information?

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How Entropy can alter a decision?

It is an interface between data and decisions.

A question to sum up the idea

Does more data bring value?

It has permitted the advances of several fields:

cryptography, neurobiology, signal processing, linguistics, bioinformatics, statistical physics, black holes, quantum computing, information retrieval, intelligence gathering, plagiarism detection, pattern recognition, anomaly detection, etc

zip files, phones, internet!

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H_X of a discrete random variable X is a measure of the amount of uncertainty associated with the value of X when only its distribution is known

What is p_i ?

It is a numerical descriptions of how likely an event is to occur

Do not mix the concepts!

Assigned probability and computed probability are different

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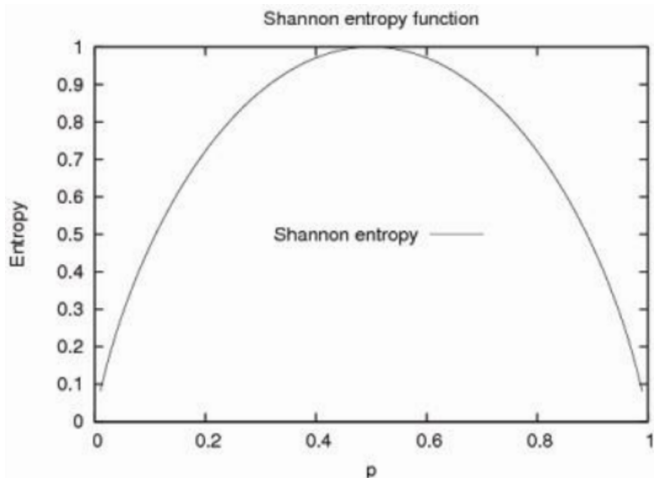
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Shannon's entropy shape

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Calculate Shannon's Entropy as a function of the number of experiments

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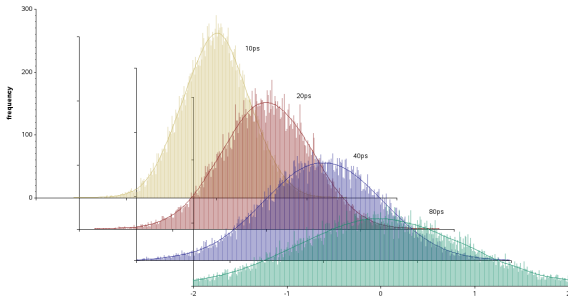
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A distribution can be the sum of multiple distributions



How to compare multiple information sources?

$$D_{KL} = \sum_i (p_i) \log\left(\frac{p_i}{q_i}\right)$$

- It is a divergence, as it is asymmetric
- It is NOT a metric

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Compare the initial discrete probability distribution with the approximate ones computed as function of the number of experiments.

Calculate Kullback–Leibler divergence for different number of experiments

Find the distribution that minimise K-L divergence